

Machine Learning 2025

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2026-01-01

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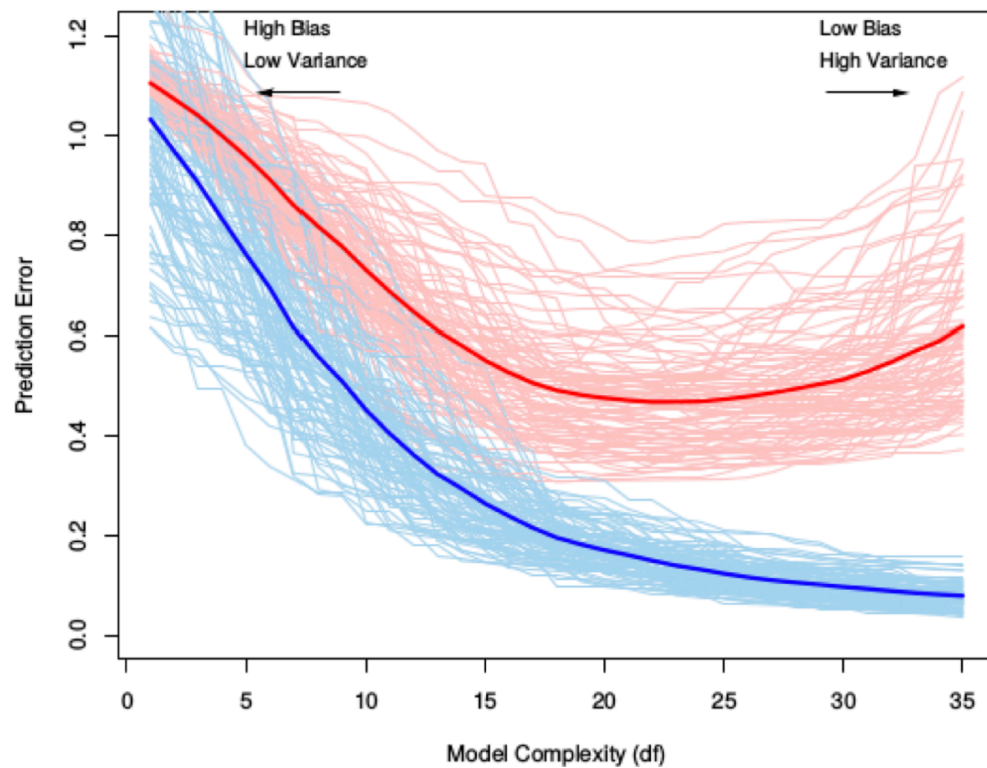


Figure 1: Training versus test error as model complexity increases.

Blue = training error, falls as complexity increases Red = test error, falls at first, then rises (U-shape) Faint curves show variability across training sets; thick curves show the typical pattern.

Left (low complexity): underfitting — the model is too rigid $\Rightarrow$ **high bias, low variance**; both **training** and **test** error are relatively high because important structure is missed.

Middle (intermediate complexity): best generalisation — as flexibility increases, bias drops and the model captures the real signal present in both train and test data $\Rightarrow$ the **test error decreases** and reaches its minimum.

Right (high complexity): overfitting — beyond a point, the model becomes too sensitive and starts fitting noise $\Rightarrow$ **variance dominates, low bias**; the **training error keeps decreasing**, but the **test error rises again** because fitted noise does not generalise well to new data.

The **U-shaped test error** reflects the sum of two competing effects:

$$\text{Test Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

As model complexity increases, Bias^2 decreases (better fit to the signal), while variance increases (greater sensitivity to the particular training set). Initially, bias reduction dominates so the **test error falls**; eventually, variance dominates so the **test error rises**.